

Topics Studied for Research

Research Topic: AI-Based Personalized Learning System

1. Topic Modeling

Topic modeling is a Natural Language Processing (NLP) technique used to discover hidden topics or themes within a collection of text documents. It helps organize, summarize, and analyze large amounts of textual data automatically.

Classical Topic Modeling Workflow

1. Text Collection
2. Text Cleaning
3. Tokenization
4. Stop Word Removal
5. Stemming or Lemmatization
6. Vectorization
7. Topic Modeling
8. Model Evaluation
9. Topic Interpretation
10. Visualization

2. Natural Language Processing (NLP)

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, and generate human language. It is widely used in applications such as chatbots, machine translation, sentiment analysis, and text summarization.

3. Tokenization

Tokenization is the process of breaking text into smaller units called tokens, such as words, sentences, or subwords. It is one of the first and most important preprocessing steps in NLP, as it prepares text data for further analysis and machine learning models.